SCHOOL OF ELECTRONICS ENGINEERING
DEPARTMENT OF MICRO AND NANO-ELECTRONICS

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COURSE CODE: MAVLD602
COURSE NAME: LOW POWER IC DESIGN
SLOT: D2 + TD2

COURSE PROJECT

Unified Power Intent (UPF) Implementation of a Simple MAC Unit

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1 Introduction

This project presents the implementation of a 16-tap FIR filter using Booth multipliers and Ripple Carry Adders (RCA). The design focuses on low-power optimization using Unified Power Format (UPF), enabling multi-voltage domain operation and power gating techniques.

2 Design Description

The Device Under Test (DUT) is a modular digital signal processing system organized as a top-level System-on-Chip (SoC) integrating computation and accumulation units. The architecture is designed to efficiently implement a multi-tap filtering operation while supporting low-power design methodologies.

At the top level, the Top SoC module coordinates data flow, control signaling, and synchronization across all sub-blocks. It maintains the input sampling sequence through a shift-register-based delay line and distributes operands to parallel computational units. The SoC also manages global control signals such as reset, clock, and enable, ensuring deterministic operation.

The Multiplier block performs signed arithmetic multiplication between delayed input samples and corresponding coefficients. A Booth encoding-based approach is employed to reduce the number of partial products, thereby improving computational efficiency and lowering switching activity. This contributes directly to reduced dynamic power consumption.

The Accumulator block aggregates the outputs of all multiplier units. It is implemented using an adder tree (or equivalent accumulation logic), enabling structured and scalable summation of multiple partial results. The design prioritizes simplicity and area efficiency while maintaining acceptable propagation delay.

Overall, the DUT follows a parallel processing architecture where multiple multipliers operate concurrently, and their outputs are combined through the accumulator. This organization enables high throughput while remaining compatible with low-power techniques such as clock gating, power gating, and multi-voltage domain partitioning.

3 RTL Implementation

3.1 Top SoC

```
module top_soc (

    input               clk,
    input               rst_n,

```

```
    input               mul_en,
    input               acc_en,
    input               acc_clear,

    input       [15:0]  a,
    input       [15:0]  b,

    output      [31:0]  result
);

    wire [31:0] product;
    wire        valid;

    multiplier u_mul (
        .clk     (clk),
        .rst_n   (rst_n),
        .enable  (mul_en),
        .a       (a),
        .b       (b),
        .product (product),
        .valid   (valid)
    );

    accumulator u_acc (
        .clk      (clk),
        .rst_n    (rst_n),
        .enable   (acc_en),
        .clear    (acc_clear),
        .data_in  (product),
        .valid_in (valid),
        .sum      (result)
    );

endmodule
```

3.2 Multiplier

```
module multiplier (
    input               clk,
    input               rst_n,
    input               enable,
```

```

    input       [15:0]  a,
    input       [15:0]  b,

    output reg [31:0]  product,
    output reg          valid
);

always @(posedge clk or negedge rst_n) begin
    if (!rst_n) begin
        product <= 32'd0;
        valid   <= 1'b0;
    end
    else begin
        product <= a * b;           // always computed
        valid   <= enable;          // only qualify output
    end
end

endmodule
```

3.3 Accumulator

```
module accumulator (
    input               clk,
    input               rst_n,
    input               enable,
    input               clear,

    input       [31:0]  data_in,
    input               valid_in,

    output reg [31:0]  sum
);

always @(posedge clk or negedge rst_n) begin
    if (!rst_n) begin
        sum <= 32'd0;
    end
    else if (clear) begin
        sum <= 32'd0;
```

```
    end
    else begin
        if (enable && valid_in)
            sum <= sum + data_in;
    end
end

endmodule
```

4 Synthesis Scripts

4.1 TCL Script (Including Constraints)

```
# -------------------------------------------------------
# Library setup (ONLY base libraries)
# -------------------------------------------------------
set_app_var target_library "/home/synopsys/installs/LIBRARIES/
    SAED14_EDK/SAED14nm_EDK_STD_RVT/liberty/nldm/base/
    saed14rvt_base_ff0p715v125c.db␣\
␣␣␣␣/home/synopsys/installs/LIBRARIES/SAED14_EDK/
    SAED14nm_EDK_STD_RVT/liberty/nldm/base/saed14rvt_base_ff0p88v125c
    .db"

set_app_var link_path "*␣\
␣␣␣␣/home/synopsys/installs/LIBRARIES/SAED14_EDK/
    SAED14nm_EDK_STD_RVT/liberty/nldm/base/
    saed14rvt_base_ff0p715v125c.db␣\
␣␣␣␣/home/synopsys/installs/LIBRARIES/SAED14_EDK/
    SAED14nm_EDK_STD_RVT/liberty/nldm/base/saed14rvt_base_ff0p88v125c
    .db"

# -------------------------------------------------------
# Read RTL
# -------------------------------------------------------
analyze -library WORK -format verilog {../rtl/top_soc.v \
                                       ../rtl/multiplier.v \
                                       ../rtl/accumulator.v \
}

elaborate top_soc
read_file -format verilog {../rtl/top_soc.v}
```

```
current_design top_soc
link
check_design

# -------------------------------------------------------
# Timing constraints
# -------------------------------------------------------
set PERIOD          5.0
set INPUT_DELAY     1.0
set OUTPUT_DELAY    1.0
set CLOCK_LATENCY   1.5
set MAX_TRANSITION  0.5

create_clock -name clk -period $PERIOD [get_ports clk]
set_clock_latency     $CLOCK_LATENCY [get_clocks clk]
set_clock_uncertainty 0.3            [get_clocks clk]
set_clock_transition  0.4            [get_clocks clk]

set INPUTPORTS  [remove_from_collection [all_inputs] [get_ports clk
    ]]
set OUTPUTPORTS [all_outputs]

set_input_delay  -clock clk -max $INPUT_DELAY  $INPUTPORTS
set_output_delay -clock clk -max $OUTPUT_DELAY $OUTPUTPORTS

set_max_transition  $MAX_TRANSITION [current_design]
set_max_capacitance 100             [current_design]
set_max_fanout      200             [current_design]

# -------------------------------------------------------
# Wire load + operating conditions
# -------------------------------------------------------
set_app_var auto_wire_load_selection false
set_wire_load_model -name 8000 -library saed14rvt_base_ff0p715v125c
set_wire_load_mode top

set_operating_conditions -library saed14rvt_base_ff0p715v125c
    ff0p715v125c

# -------------------------------------------------------
# Compile
```

```
# -------------------------------------------------------
compile_ultra

# -------------------------------------------------------
# Reports
# -------------------------------------------------------
report_timing
report_area    > ./reports/top_soc_area.rpt
report_power   > ./reports/top_soc_power.rpt
report_timing  > ./reports/top_soc_timing.rpt
report_qor     > ./reports/top_soc_qor.rpt

# -------------------------------------------------------
# Outputs
# -------------------------------------------------------
write -format verilog -hierarchy -output ./reports/top_soc_netlist.v
write_sdc ./reports/top_soc_constraints.sdc
```

5 Power Analysis Before UPF

The power analysis of the design was performed at a global operating voltage of **0.715 V**. The results are summarized below:

- Cell Internal Power: **73.37** μ**W** (33%)
- Net Switching Power: **145.97** μ**W** (67%)
- Total Dynamic Power: **219.34** μ**W**
- Cell Leakage Power: **22.41** μ**W**

The total power consumption of the design is:

$$\textbf{Total Power = 0.2417 mW} \tag{1}$$

5.1 Power Distribution Analysis

The distribution of power across different components is as follows:

- Combinational logic: **82.05%**
- Clock network: **15.28%**

- Registers: **2.67%**
- Other components (I/O, memory, black-box): **0%**

5.2 Power Report Visualization

```
****************************************
Report : power
        -analysis_effort low
Design : top_soc
Version: V-2023.12-SP4
Date   : Fri Apr 17 12:49:43 2026
****************************************


Library(s) Used:

    saed14rvt_dlvl_ff0p715v125c_i0p88v (File: /home/synopsys/installs/LIBRARIES/SAED14_EDK/SAED14nm_EDK_STD_RVT/liberty/nldm/dlvl/saed14rvt_dlvl_ff0p715v125c_i0p88v.db)
    saed14rvt_ulvl_ff0p88v125c_i0p715v (File: /home/synopsys/installs/LIBRARIES/SAED14_EDK/SAED14nm_EDK_STD_RVT/liberty/nldm/ulvl/saed14rvt_ulvl_ff0p88v125c_i0p715v.db)
    saed14rvt_base_ff0p715v125c (File: /home/synopsys/installs/LIBRARIES/SAED14_EDK/SAED14nm_EDK_STD_RVT/liberty/nldm/base/saed14rvt_base_ff0p715v125c.db)


Global Operating Voltage = 0.715
Power-specific unit information :
    Voltage Units = 1V
    Capacitance Units = 1.000000pf
    Time Units = 1ns
    Dynamic Power Units = 1mW    (derived from V,C,T units)
    Leakage Power Units = 1pW


Attributes
----------
i - Including register clock pin internal power


  Cell Internal Power  =  73.3704 uW   (33%)
  Net Switching Power  = 145.9685 uW   (67%)
                         ---------
Total Dynamic Power    = 219.3389 uW  (100%)

Cell Leakage Power     =  22.4069 uW


                 Internal         Switching           Leakage            Total
Power Group      Power            Power               Power              Power   (   %    )  Attrs
--------------------------------------------------------------------------------------------------
io_pad             0.0000           0.0000            0.0000            0.0000  (   0.00%)
memory             0.0000           0.0000            0.0000            0.0000  (   0.00%)
black_box          0.0000           0.0000            0.0000            0.0000  (   0.00%)
clock_network  3.6945e-02           0.0000            0.0000        3.6945e-02  (  15.28%)  i
register       1.3957e-03        3.5948e-03        1.4526e+06        6.4431e-03  (   2.67%)
sequential         0.0000           0.0000            0.0000            0.0000  (   0.00%)
conbinational  3.5030e-02           0.1424        2.0954e+07            0.1984  (  82.05%)
--------------------------------------------------------------------------------------------------
Total          7.3370e-02 mW        0.1460 mW     2.2407e+07 pW        0.2417 mW
1
```

Figure 1: Power Report Before UPF Implementation

5.3 Observation

The design is dominated by switching power, with net switching contributing the largest portion of total power. This indicates that techniques such as voltage scaling, reduction of switching activity, and power gating can be effectively used to reduce overall power consumption.

6 UPF Integration

To enable low-power operation, the design is partitioned into multiple power domains using the Unified Power Format (UPF). The architecture separates high-switching arithmetic blocks from control and accumulation logic to allow independent power management.

The DUT is divided into two primary power domains:

- **PD_CORE**: Contains the Top SoC and accumulator logic, operating at a higher voltage of **0.85 V** to maintain timing performance.
- **PD_MUL**: Contains the multiplier units, operating at a reduced voltage of **0.7 V** to minimize dynamic and leakage power.

6.1 Supply Network Definition

Separate supply ports and nets are defined for each domain. Dedicated supply sets are created to associate power and ground connections for both PD_CORE and PD_MUL. This enables proper modeling of multi-voltage operation within the design.

6.2 Power Domain Partitioning

The multiplier instances are explicitly grouped into the PD_MUL domain, while the remaining logic resides in PD_CORE. This partitioning isolates the most power-intensive components, allowing targeted optimization.

6.3 Power Gating using Power Switch

A power switch is introduced for the PD_MUL domain. The switch is controlled by a dedicated signal, allowing the multiplier block to be turned OFF during idle periods. This effectively reduces leakage power by disconnecting the supply when the block is inactive.

6.4 Isolation Strategy

Isolation cells are inserted at the boundary of PD_MUL outputs. When the multiplier domain is powered down, these cells clamp the outputs to a known logic value, preventing undefined signal propagation into the active PD_CORE domain.

6.5 Level Shifter Insertion

Since the two domains operate at different voltage levels, level shifters are used at domain interfaces. These ensure reliable signal transfer between PD_MUL (low voltage) and PD_CORE (higher voltage), avoiding signal integrity issues.

6.6 Power State Definition

Power states are defined for each supply set to describe valid operating conditions. Both domains are active during normal operation, while PD_MUL can be selectively powered down based on control signals to achieve power savings.

6.7 Summary

The UPF-based implementation enables:

- Multi-voltage domain operation
- Leakage reduction through power gating
- Safe inter-domain communication via isolation and level shifting

This structured power intent allows the design to achieve significant power savings while maintaining functional correctness and timing reliability.

7 Modified RTL for UPF

7.1 Modified Top SoC

```

module top_soc (

    input               clk,
    input               rst_n,

    // control
    input               mul_en,
    input               acc_en,
    input               acc_clear,

    // inputs
    input       [15:0]  a,
    input       [15:0]  b,

    // output
    output   reg  [31:0]  result
);

    wire [31:0] product;
```

```
    wire        valid;


    multiplier u_mul (
        .clk     (clk),
        .rst_n   (rst_n),
        .enable  (mul_en),
        .a       (a),
        .b       (b),
        .product (product),
        .valid   (valid)
    );


    accumulator u_acc (
        .clk      (clk),
        .rst_n    (rst_n),
        .enable   (acc_en),
        .clear    (acc_clear),
        .data_in  (product),
        .valid_in (valid),
        .sum      (result)
    );

endmodule
```

8 UPF Script

```
############################################################
# SUPPLY NETS
############################################################

create_supply_net VSS
create_supply_net VDDL
create_supply_net VDDH

############################################################
# SUPPLY SETS
############################################################

```

```
create_supply_set ss_high \
    -function {power VDDH} \
    -function {ground VSS}

create_supply_set ss_low \
    -function {power VDDL} \
    -function {ground VSS}

############################################################
# POWER DOMAINS
############################################################

# Top domain (High VDD)
create_power_domain PD_TOP \
    -supply {primary ss_high}

# Multiplier (Low VDD)
create_power_domain PD_MUL \
    -elements u_mul \
    -supply {primary ss_low}

# Accumulator (Low VDD)
create_power_domain PD_ACC \
    -elements u_acc \
    -supply {primary ss_low}

############################################################
# SUPPLY PORTS (TOP LEVEL)
############################################################

create_supply_port VDDH -domain PD_TOP -direction in
create_supply_port VDDL -domain PD_TOP -direction in
create_supply_port VSS  -domain PD_TOP -direction in

############################################################
# CONNECT SUPPLIES
############################################################

connect_supply_net VDDH -ports VDDH
connect_supply_net VDDL -ports VDDL
connect_supply_net VSS  -ports VSS
```

```

##############################################################
# LEVEL SHIFTERS - G4 PLACEMENT (ALL IN PARENT = PD_TOP)
#
# G4 style: both input and output level shifters for
# PD_MUL and PD_ACC are placed in the parent domain (PD_TOP).
# This means ALL LS cells are powered by VDDH (ss_high).
#
#   PD_TOP        [dlvl LS]      > PD_MUL       >[ulvl LS]      > PD_TOP
#   PD_TOP        [dlvl LS]      > PD_ACC       >[ulvl LS]      > PD_TOP
##############################################################

# SoC -> MUL (High->Low) : LS placed OUTSIDE MUL, in PD_TOP (parent)
set_level_shifter LS_SOC_MUL \
    -domain PD_MUL \
    -applies_to inputs \
    -rule high_to_low \
    -location parent

# MUL -> SoC (Low->High) : LS placed OUTSIDE MUL, in PD_TOP (parent)
set_level_shifter LS_MUL_SOC \
    -domain PD_MUL \
    -applies_to outputs \
    -rule low_to_high \
    -location parent

# SoC -> ACC (High->Low) : LS placed OUTSIDE ACC, in PD_TOP (parent)
set_level_shifter LS_SOC_ACC \
    -domain PD_ACC \
    -applies_to inputs \
    -rule high_to_low \
    -location parent

# ACC -> SoC (Low->High) : LS placed OUTSIDE ACC, in PD_TOP (parent)
set_level_shifter LS_ACC_SOC \
    -domain PD_ACC \
    -applies_to outputs \
    -rule low_to_high \
    -location parent
```

9 Synthesis Script with UPF

```
set_app_var target_library "/home/synopsys/installs/LIBRARIES/SAED14_EDK/SAED14nm_EDK_STD_RVT/liberty/nldm/base/saed14rvt_base_ff0p715v125c.db␣\
␣␣␣␣/home/synopsys/installs/LIBRARIES/SAED14_EDK/SAED14nm_EDK_STD_RVT/liberty/nldm/base/saed14rvt_base_ff0p88v125c.db␣\
␣␣␣␣/home/synopsys/installs/LIBRARIES/SAED14_EDK/SAED14nm_EDK_STD_RVT/liberty/nldm/ulvl/saed14rvt_ulvl_ff0p88v125c_i0p715v.db␣\
␣␣␣␣/home/synopsys/installs/LIBRARIES/SAED14_EDK/SAED14nm_EDK_STD_RVT/liberty/nldm/dlvl/saed14rvt_dlvl_ff0p715v125c_i0p88v.db"
set_app_var link_path "*␣\
␣␣␣␣/home/synopsys/installs/LIBRARIES/SAED14_EDK/SAED14nm_EDK_STD_RVT/liberty/nldm/base/saed14rvt_base_ff0p715v125c.db␣\
␣␣␣␣/home/synopsys/installs/LIBRARIES/SAED14_EDK/SAED14nm_EDK_STD_RVT/liberty/nldm/base/saed14rvt_base_ff0p88v125c.db␣\
␣␣␣␣/home/synopsys/installs/LIBRARIES/SAED14_EDK/SAED14nm_EDK_STD_RVT/liberty/nldm/ulvl/saed14rvt_ulvl_ff0p88v125c_i0p715v.db␣\
␣␣␣␣/home/synopsys/installs/LIBRARIES/SAED14_EDK/SAED14nm_EDK_STD_RVT/liberty/nldm/dlvl/saed14rvt_dlvl_ff0p715v125c_i0p88v.db"

analyze -library WORK -format verilog {../rtl/top_soc.v \
                                       ../rtl/multiplier.v \
                                       ../rtl/accumulator.v \
}

elaborate top_soc
read_file -format verilog {../rtl/top_soc.v}
current_design top_soc
link
check_design

# G4 placement UPF
load_upf design_g4.upf
```

```tcl

# -------------------------------------------------------
# Voltage annotation (FIXED from original dc_ls.tcl)
# PD_TOP  -> VDDH = 0.88V
# PD_MUL  -> VDDL = 0.715V
# PD_ACC  -> VDDL = 0.715V
# -------------------------------------------------------
set_voltage 0.715 -object_list VDDL
set_voltage 0.88  -object_list VDDH
set_voltage 0.0   -object_list VSS

set_voltage 0.88  -object_list PD_TOP.primary.power
set_voltage 0.0   -object_list PD_TOP.primary.ground
set_voltage 0.715 -object_list PD_MUL.primary.power
set_voltage 0.0   -object_list PD_MUL.primary.ground
set_voltage 0.715 -object_list PD_ACC.primary.power
set_voltage 0.0   -object_list PD_ACC.primary.ground

check_mv_design

# -------------------------------------------------------
# Timing constraints
# -------------------------------------------------------
set PERIOD       5.0
set INPUT_DELAY  1.0
set OUTPUT_DELAY 1.0
set CLOCK_LATENCY 1.5
set MAX_TRANSITION 0.5

create_clock -name clk -period $PERIOD [get_ports clk]
set_clock_latency    $CLOCK_LATENCY [get_clocks clk]
set_clock_uncertainty 0.3           [get_clocks clk]
set_clock_transition  0.4           [get_clocks clk]

set INPUTPORTS  [remove_from_collection [all_inputs] [get_ports clk
    ]]
set OUTPUTPORTS [all_outputs]
set_input_delay  -clock clk -max $INPUT_DELAY  $INPUTPORTS
set_output_delay -clock clk -max $OUTPUT_DELAY $OUTPUTPORTS

set_max_transition  $MAX_TRANSITION [current_design]
```

```
set_max_capacitance 100             [current_design]
set_max_fanout      200             [current_design]

set_app_var auto_wire_load_selection false
set_wire_load_model -name 8000 -library saed14rvt_base_ff0p715v125c
set_wire_load_mode top
set_operating_conditions -library saed14rvt_base_ff0p715v125c
   ff0p715v125c

compile_ultra

report_timing
report_area    > ./reports/top_soc_g4_area.rpt
report_power   > ./reports/top_soc_g4_power.rpt
report_timing  > ./reports/top_soc_g4_timing.rpt
report_qor     > ./reports/top_soc_g4_qor.rpt
write -format verilog -hierarchy -output ./reports/
   top_soc_g4_netlist.v
write_sdc ./reports/top_soc_g4_constraints.sdc
```

10 Power Analysis After UPF

The power analysis of the UPF-enabled design was performed at a global operating voltage of **0.715 V**. The results after applying multi-voltage domain optimization and power intent are summarized below:

- Cell Internal Power: **73.95** μ**W** (40%)
- Net Switching Power: **112.57** μ**W** (60%)
- Total Dynamic Power: **186.52** μ**W**
- Cell Leakage Power: **13.53** μ**W**

The total power consumption of the design is:

$$\textbf{Total Power = 0.2000 mW} \tag{2}$$

10.1 Power Distribution Analysis

The distribution of power across different components is as follows:

- Combinational logic: **75.76%**

- Clock network: **20.66%**
- Registers: **3.58%**
- Other components (I/O, memory, black-box): **0%**

10.2 Power Report Visualization

```
****************************************
Report : power
        -analysis_effort low
Design : top_soc
Version: V-2023.12-SP4
Date   : Fri Apr 17 13:22:34 2026
****************************************


Library(s) Used:

    saed14rvt_base_ff0p715v125c (File: /home/synopsys/installs/LIBRARIES/SAED14_EDK/SAED14nm_EDK_STD_RVT/liberty/nldm/base/saed14rvt_base_ff0p715v125c.db)


Operating Conditions: ff0p715v125c   Library: saed14rvt_base_ff0p715v125c
Wire Load Model Mode: top

Design        Wire Load Model            Library
------------------------------------------------
top_soc               8000              saed14rvt_base_ff0p715v125c


Global Operating Voltage = 0.715
Power-specific unit information :
    Voltage Units = 1V
    Capacitance Units = 1.000000pf
    Time Units = 1ns
    Dynamic Power Units = 1mW    (derived from V,C,T units)
    Leakage Power Units = 1pW


Attributes
----------
i - Including register clock pin internal power


  Cell Internal Power  =  73.9476 uW   (40%)
  Net Switching Power  = 112.5698 uW   (60%)
                         ---------
Total Dynamic Power    = 186.5175 uW  (100%)

Cell Leakage Power     =  13.5284 uW


                 Internal         Switching           Leakage            Total
Power Group      Power            Power               Power              Power   (   %    )  Attrs
--------------------------------------------------------------------------------------------------
io_pad             0.0000           0.0000            0.0000            0.0000  (   0.00%)
memory             0.0000           0.0000            0.0000            0.0000  (   0.00%)
black_box          0.0000           0.0000            0.0000            0.0000  (   0.00%)
clock_network  4.1334e-02           0.0000            0.0000        4.1334e-02  (  20.66%)  i
register       2.2590e-03        3.1625e-03        1.7311e+06        7.1526e-03  (   3.58%)
sequential         0.0000           0.0000            0.0000            0.0000  (   0.00%)
combinational  3.0355e-02           0.1094        1.1797e+07            0.1516  (  75.76%)
--------------------------------------------------------------------------------------------------
Total          7.3948e-02 mW        0.1126 mW     1.3528e+07 pW         0.2000 mW
1
```

Figure 2: Power Report after UPF Implementation

10.3 Observation

After UPF implementation, a noticeable reduction in both dynamic and leakage power is observed. The decrease in switching power indicates effective voltage scaling and reduced activity in power-gated domains. Leakage power is also reduced due to power gating of the multiplier domain.

The design remains predominantly combinational-power driven, but with improved efficiency compared to the pre-UPF implementation.

11 Outputs and Screenshots

This section presents the implementation results of the UPF-based design, including power domain visualization, level shifter insertion, and tool-generated reports obtained after synthesis.

11.1 Level Shifter Report

The `report_level_shifter` output confirms correct insertion of level shifters across voltage domains. A total of **68 level shifters** were inserted in the design, with **no violations** reported.

All level shifters operate between high-voltage (VDDH = 0.88V) and low-voltage (VDDL = 0.71V) domains, ensuring safe signal transfer. The absence of violations indicates proper UPF constraint specification and successful implementation of level shifting strategies.

```
dc_shell> report_level_shifter

****************************************
Report : level_shifter
Design : top_soc
Version: V-2023.12-SP4
Date   : Fri Apr 17 12:52:11 2026
****************************************

-------------------------------------------------------------------------------------------------------------------------------------------------------------

 Power Domain : PD_TOP

 ************************************************************
 Level shifter summary for power domain
 ************************************************************
 No. of violating level shifters - dont_touch    : 0
 No. of violating level shifters - no dont_touch : 0
 No. of level shifters in domain                 : 68
 ************************************************************

=============================================================================================================================================================
 Level Shifter       Reference       Input P/G      Output P/G     Main P/G       Violation   Reason                 Voltage             Type
                                     Net Info(Volt) Net Info(Volt) Net Info(Dir.)
=============================================================================================================================================================
 b[6]_UPF_LS         SAEDRVT14_LVLDBUF_IY2V1_2
                                     VDDH(0.88)     VDDL(0.71)     VDDH(---)      FALSE       -                      (0.585,0.88)        HL
                                     VSS(0.00)      VSS(0.00)      ---
 b[1]_UPF_LS         SAEDRVT14_LVLDBUF_IY2V1_2
                                     VDDH(0.88)     VDDL(0.71)     VDDH(---)      FALSE       -                      (0.585,0.88)        HL
                                     VSS(0.00)      VSS(0.00)      ---
 rst_n_UPF_LS        SAEDRVT14_LVLDBUF_IY2V1_1
                                     VDDH(0.88)     VDDL(0.71)     VDDH(---)      FALSE       -                      (0.585,0.88)        HL
                                     VSS(0.00)      VSS(0.00)      ---
 enable_1_UPF_LS     SAEDRVT14_LVLDBUF_IY2V1_1
                                     VDDH(0.88)     VDDL(0.71)     VDDH(---)      FALSE       -                      (0.585,0.88)        HL
                                     VSS(0.00)      VSS(0.00)      ---
 enable_UPF_LS       SAEDRVT14_LVLDBUF_IY2V1_1
                                     VDDH(0.88)     VDDL(0.71)     VDDH(---)      FALSE       -                      (0.585,0.88)        HL
                                     VSS(0.00)      VSS(0.00)      ---
```

Figure 3: Level Shifter Report from Design Compiler

11.2 UPF Power Domain Schematic

The schematic view illustrates the hierarchical organization of power domains within the design. The top-level domain (**PD_TOP**) encapsulates two subdomains:

- **PD_ACC**: Accumulator domain
- **PD_MUL**: Multiplier domain

The diagram clearly shows domain boundaries and interconnections, highlighting how different functional blocks are partitioned for power optimization.

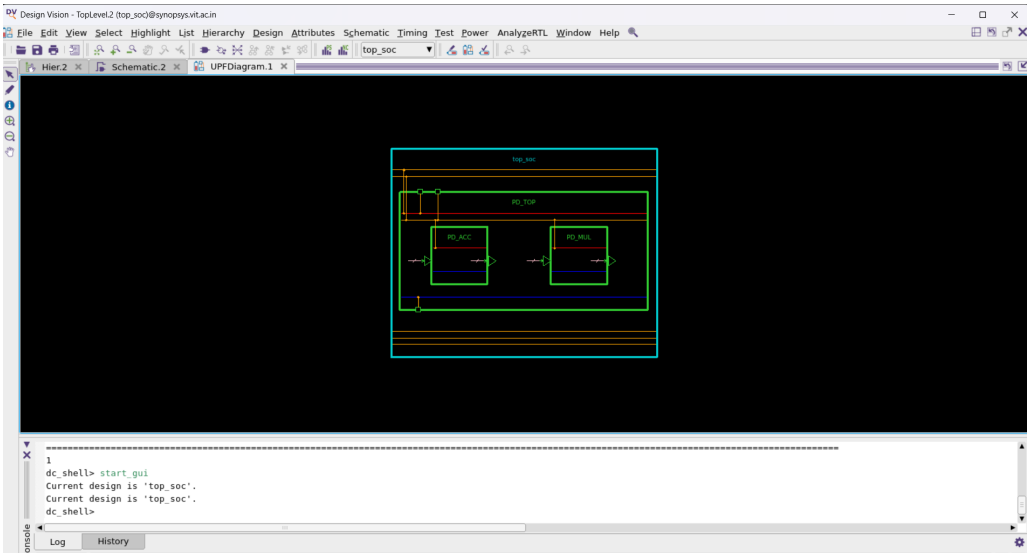

Figure 4: UPF Power Domain Schematic

11.3 Visual UPF Hierarchy

The Visual UPF tool provides a structured representation of the design hierarchy, including supply nets, supply sets, and power domains. It confirms the correct mapping of:

- Supply nets: `VDDH`, `VDDL`, and `VSS`
- Power domains: **PD_TOP**, **PD_MUL**, and **PD_ACC**
- Level shifter strategies between domains

This visualization validates that the UPF intent has been correctly interpreted and applied during synthesis.

11.4 Summary of Results

The outputs confirm that:

- Multi-voltage domain implementation is successful
- Level shifters are correctly inserted with zero violations
- Power domain partitioning is properly realized

These results validate the correctness of the UPF-based low-power design methodology.

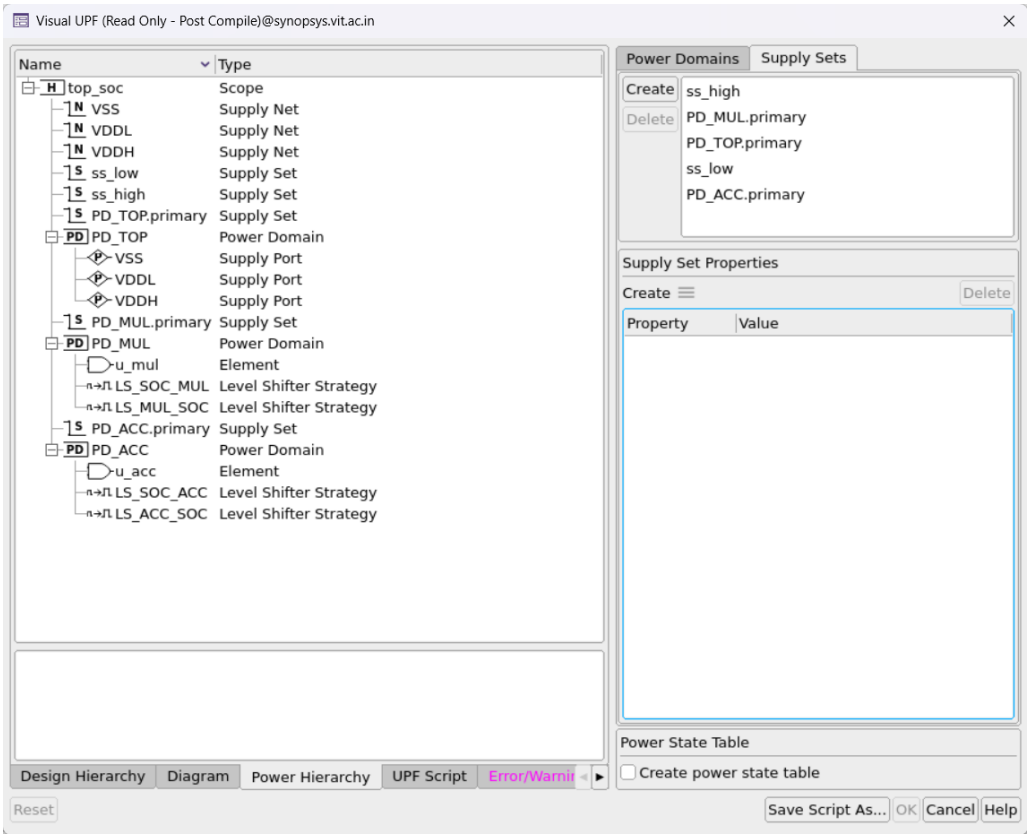

Figure 5: Visual UPF Hierarchy and Power Domains

12 Results and Analysis

The implementation of UPF-based low-power techniques resulted in a measurable improvement in overall power efficiency of the design. A comparative analysis of the design before and after UPF integration highlights the effectiveness of multi-voltage domain partitioning and power-aware optimizations.

12.1 Power Comparison

Metric	Before UPF	After UPF
Total Dynamic Power	219.34 μW	186.52 μW
Cell Leakage Power	22.41 μW	13.53 μW
Total Power	0.2417 mW	0.2000 mW

12.2 Power Reduction Analysis

The total power consumption is reduced from **0.2417 mW** to **0.2000 mW**, resulting in an approximate reduction of:

$$\textbf{Power Reduction} = \textbf{17.25\%} \tag{3}$$

Leakage power shows a significant reduction due to power gating:

$$\textbf{Leakage Reduction = 39.63\%} \tag{4}$$

Dynamic power is also reduced due to voltage scaling and reduced switching activity in the multiplier domain.

12.3 Impact of UPF Techniques

The observed improvements can be attributed to the following:

- **Multi-Voltage Domains:** The multiplier domain operates at a lower voltage, reducing dynamic power consumption.
- **Power Gating:** The PD_MUL domain can be turned off during idle periods, significantly reducing leakage power.
- **Level Shifters:** A total of **68 level shifters** were inserted to safely interface signals between voltage domains without any violations.
- **Isolation Cells:** Ensure signal integrity when the low-power domain is switched off.

12.4 Design Efficiency Observation

The power distribution indicates that combinational logic remains the dominant contributor to total power consumption. However, its overall impact is reduced after UPF implementation due to lower operating voltage in critical blocks.

Additionally, the clock network contribution slightly increases in percentage due to the relative reduction in combinational power, not due to increased clock activity.

12.5 Conclusion of Analysis

The results demonstrate that UPF-driven design methodologies effectively reduce both dynamic and leakage power without compromising functionality. The integration of power domains, level shifters, and power gating enables a scalable and efficient low-power design suitable for modern digital systems.